Unit 1 Review

> Everything Unit 1 taught, one small example each, to use while you build Rock Paper Scissors.

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1. Show, store, and ask

print() shows text in the terminal. **A variable** stores a value. `=` sets something equal. Use clear names.

```
print("Hello, world!")
color = "blue"
print(color)
```

input() stores what the user types. **An f-string** (`f` before the quote) fills in each `{name}`.

```
pet = input("What's your pet's name? ")
snack = input("What does it eat? ")
print(f"I bet {pet} loves {snack}.")
```

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2. Numbers and comments

int() turns text like `"12"` into a number for math or `>=`. **A comment** starts with `#`, and Python skips it.

```
# Design: ask for an age, then show next year's age.
age = int(input("How old are you? "))
next_year = age + 1
print(f"Next year you'll be {next_year}.")
```

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3. Decisions

if runs its indented lines only when the condition is true.

`=` sets something equal. `==` asks if it is equal. `color = "blue"` sets it. `color == "blue"` asks.

```
answer = input("Do you like soup? yes/no: ")
if answer == "yes":
    print("Soup for you!")
```

if / elif / else: Python runs the first branch that is true, then stops. `else` catches everything else.

```
temp = int(input("What's the temperature? "))
if temp >= 25:
    print("Wear shorts.")
elif temp >= 10:
    print("Wear a jacket.")
else:
    print("Wear a coat.")
```

and needs **both** sides true. **or** needs **at least one**. Write the full comparison on each side.

```
weather = input("Weather? ")
day = input("Day? ")
if weather == "sunny" and day == "saturday":
    print("Beach day!")
if day == "saturday" or day == "sunday":
    print("No school!")
```

A nested if is an if inside an if. The second question is asked only on the `yes` path.

```
hungry = input("Are you hungry? yes/no: ")
if hungry == "yes":
    sweet = input("Want something sweet? yes/no: ")
    if sweet == "yes":
        print("Try a cookie.")
    else:
        print("Try a pretzel.")
```

Test every path. Run once per branch, then try the input most likely to break it. Type in lowercase: `Yes` is not `"yes"`.